

**JAYPEE INSTITUTE OF INFORMATION
TECHNOLOGY,**

NOIDA

B-TECH SEMESTER-2



PBL Report

TOPIC:

Digital Habit Tracker

BATCH – F-17

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1. Introduction

- **Background and Context:**

In today's fast-paced digital world, maintaining consistency in daily routines is challenging. People often struggle to build productive habits such as exercising, studying, or maintaining a healthy lifestyle. Traditional tracking methods like notebooks are inefficient and lack motivation features.

The **Digital Habit Tracker** is a console-based C++ application designed to help users build, track, and maintain habits effectively. It uses Object-Oriented Programming principles and provides a visually enhanced terminal interface.

- **Problem Statement:**

In today's busy and distraction-filled lifestyle, individuals often struggle to maintain consistency in their daily and weekly habits. While people aim to improve productivity, health, and personal growth, they frequently fail due to the lack of an effective system to track and monitor their progress. Traditional methods such as writing in notebooks or relying on memory are inefficient, prone to errors, and do not provide any form of motivation or performance analysis. Additionally, users find it difficult to manage multiple habits simultaneously and assess their consistency over time. There is also a lack of reward-based systems that encourage users to stay committed. Therefore, there is a need for a digital solution that not only tracks habits efficiently but also provide progress visualization, performance insights, and motivational features like streaks, points, and badges. The Digital Habit Tracker aims to address these challenges by offering a structured, interactive, and user-friendly system for effective habit management.

- Project Objectives:

- To develop a habit tracking system using C++
- To implement Object-Oriented Programming concepts
- To allow users to:
 - Add and manage habits
 - Track daily/weekly progress
 - Maintain streaks and points
- To provide motivation using:
 - Badges (Bronze, Silver, Gold, Legend)
 - Leaderboard system
- To store and retrieve data using file handling.

- Scope of the Project:

The scope of the Digital Habit Tracker project is focused on developing a console-based application that helps users efficiently create, manage, and monitor their daily and weekly habits. The system allows users to add, delete, search, and mark habits as completed while tracking important metrics such as streaks, points, and completion status. It also provides performance insights through reports, a dashboard, and a leaderboard, making habit tracking more engaging and motivating. The application uses file handling to ensure data persistence, allowing users to retain their progress across multiple sessions. Although the current implementation is limited to a single-user, console-based environment, it establishes a strong foundation that can be extended in the future to include graphical user interfaces, mobile application development, multi-user support, cloud storage, and advanced analytics. Thus, the project not only fulfils basic habit tracking needs but also has significant potential for future enhancements and real-world applications.

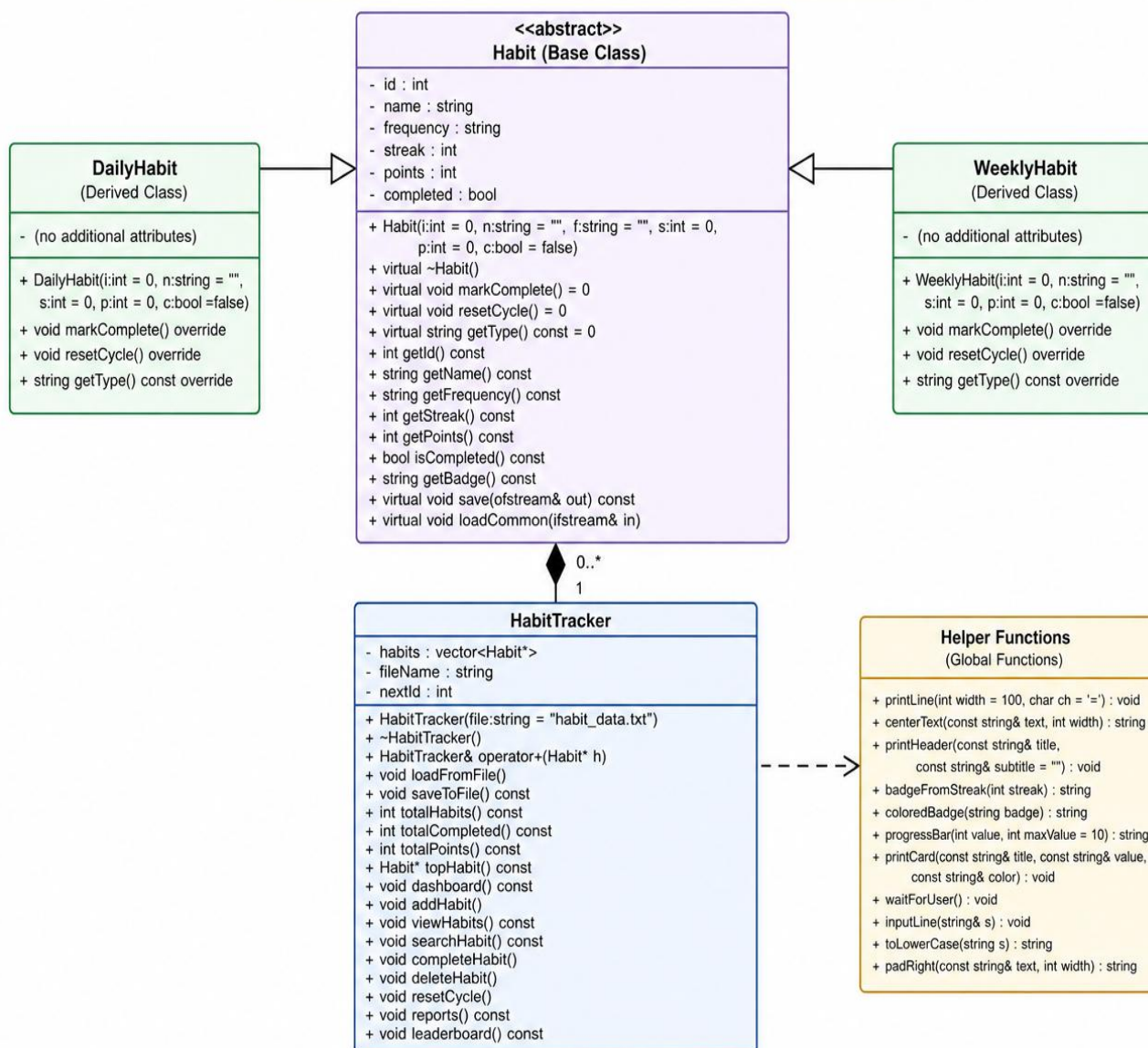
2.Design of the project

- Flowchart depicting functioning of the project:



- Class diagram:

DIGITAL HABIT TRACKER – CLASS DIAGRAM



3. Conclusion

- **Summary:**

The Digital Habit Tracker project successfully implements a complete habit management system using C++. It demonstrates:

- Strong use of OOP concepts
- Efficient data handling using STL
- Persistent storage using file handling
- Interactive and user-friendly console interface

The application allows users to perform various operations such as adding new habits, deleting existing ones, searching for specific habits, and marking habits as completed. Each habit is tracked using key parameters such as streak count, points earned, and completion status. The system differentiates between daily and weekly habits, enabling flexibility in tracking different types of routines. The project not only solves real-world problems but also enhances programming skills.

- **Future Work and Recommendations:**

The system can be improved by:

1. Adding Graphical User Interface (GUI)
2. Implementing Database (MySQL / Firebase)
3. Adding Notifications & Reminders
4. Creating Mobile Application Version
5. Adding Data Visualization (Graphs)
6. Implementing User Authentication System
7. Integrating AI-based habit suggestions

4. References

- <https://www.cplusplus.com>
- <https://www.geeksforgeeks.org>
- <https://stackoverflow.com>
- C++ STL Documentation
- SDF PPTs
- YouTube Resources